

27 Nov 2023

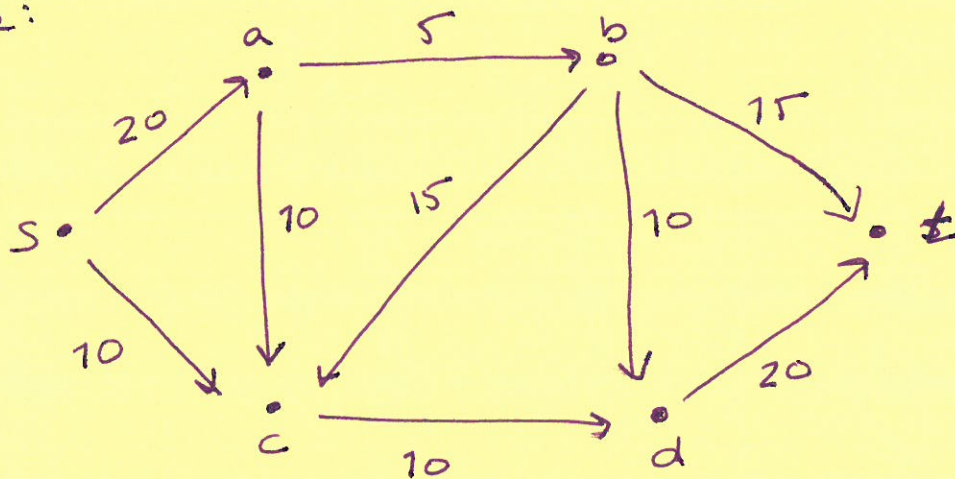
Max Flow / Min Cut

input: directed graph $G = (V, E)$
 source $s \in V$
 sink $t \in V$
 capacity fn $c: E \rightarrow \mathbb{R}_{\geq 0}$

output: max flow from s to $t \rightarrow$ Max Flow Problem

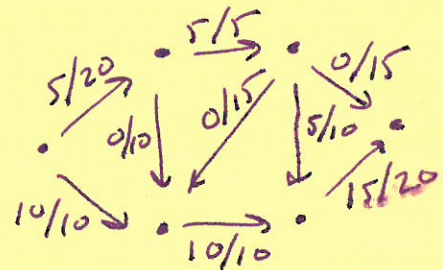
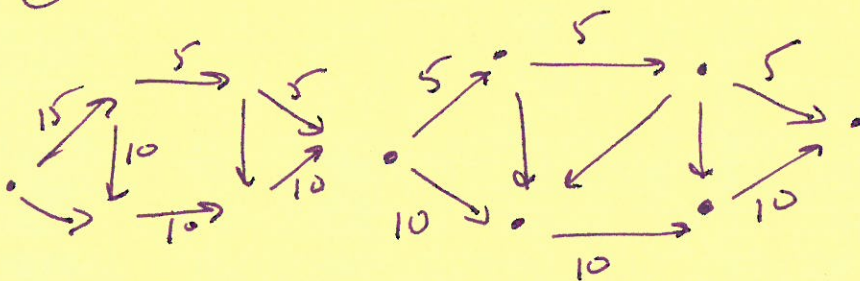
min cut separating s & $t \rightarrow$ Min cut Problem

Example:



Key:
 $\xrightarrow{c}$
 is capacity of edge
 $\xrightarrow{f/c}$
 f = flow
 c = capacity

① What is the value of a max flow? 15



HINT: Start w/ any flow, then ask yourself "can I do better"?

①

Flows

An (s, t) -flow is a function $f: E \rightarrow \mathbb{R}_{\geq 0}$ that satisfies conservation (of flow) ~~everywhere~~ at every edge in $V \setminus \{t, s\}$: i.e., $\forall v \in V \setminus \{t, s\}$,

$$\sum_{u \in V} f(u \rightarrow v) = \sum_{w \in V} f(v \rightarrow w)$$

"flow in" = "flow out"

where

$$f(x \rightarrow y) := 0, \text{ if } x \rightarrow y \notin E$$

We say f is feasible if $\forall e \in E$,

$$0 \leq f(e) \leq c(e).$$

Def: The value of flow f is

$$|f| := \sum_{w \in V} f(s \rightarrow w) - \sum_{u \in V} f(u \rightarrow s).$$

Max Flow Problem:

Find the maximum value of any feasible flow from s to t .

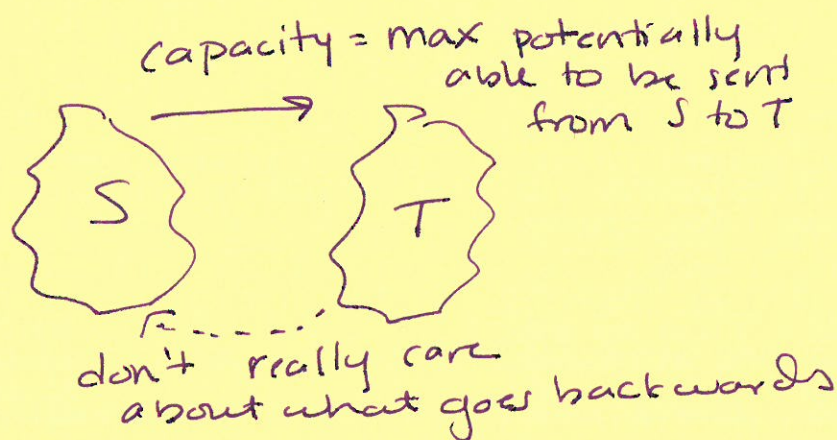
(You can also find a max flow)

Cuts

An (s,t) -cut is a partition of the vertices $V = S \sqcup T$ such that $s \in S$ and $t \in T$.

The capacity of a cut (S,T) is:

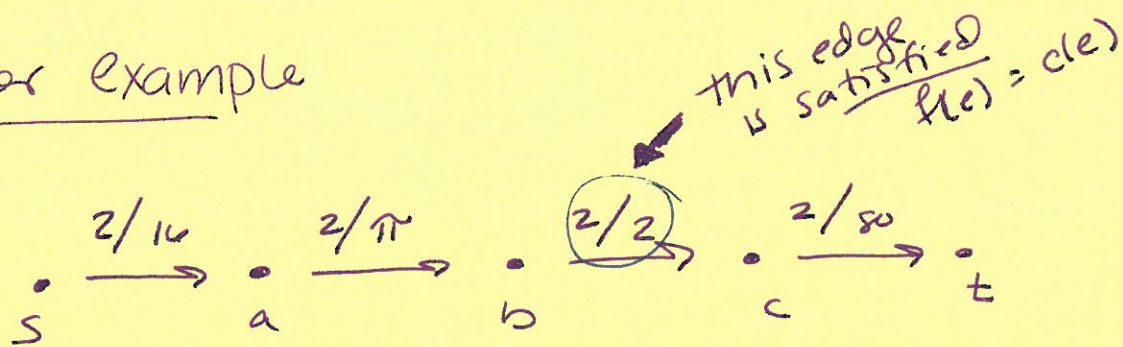
$$\|S,T\| := \sum_{v \in S} \sum_{w \in T} c(v \rightarrow w)$$



Min Cut Problem:

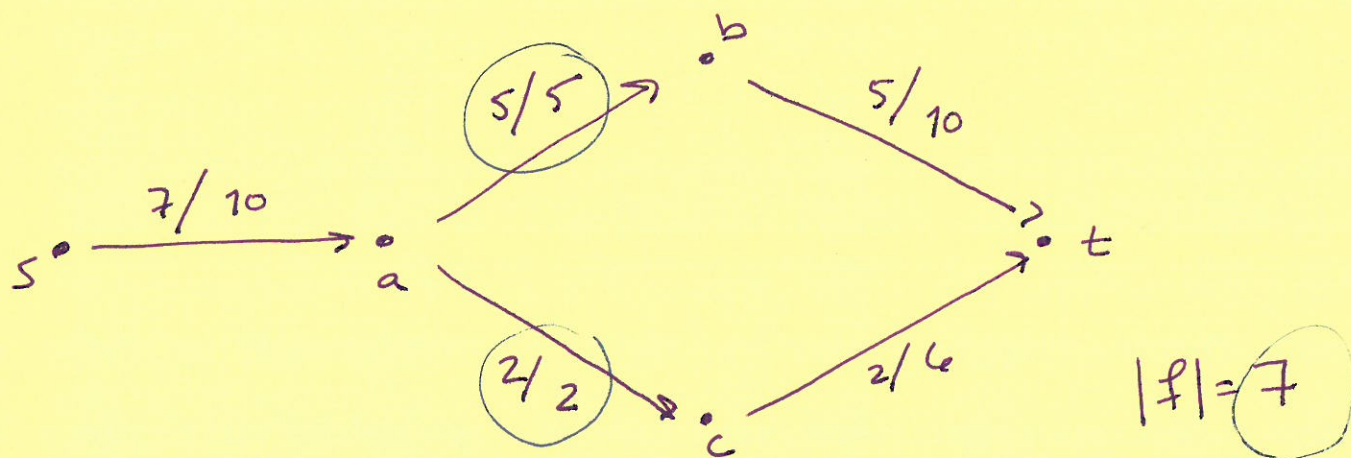
Find the minimum capacity of any (s,t) -cut.
(+ what is a cut that realizes that minimum)

Another example



• If I just have a path, the max flow = the ^{smallest} capacity of the edges
in function form:

$$f(e) = 2, \quad \forall e \in E$$



Min cut:

$$S = \{s, a\}$$

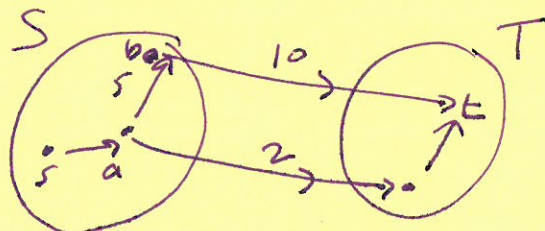
$$T = \{t, b, c\}$$

$$||S, T|| = 5 + 2 = 7$$

A cut:

$$S = \{s, a, b\}$$

$$T = \{t, c\}$$



$$||S, T|| = 12$$

(4)